

VNBANK-LOD: A Five-Star Linked Open Data Dataset of Vietnamese Banks

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Abstract

Public information about Vietnamese banks is published as HTML pages, PDF reports and spreadsheets that machines cannot join or query, which places it at one or two stars on the five-star open data scheme. This report presents VNBANK-LOD, a Linked Open Data application for the Vietnamese banking domain that follows the full five-star path. We design an OWL 2 ontology with four core classes, five defined classes, cardinality, inverse and disjointness axioms, and an alignment to schema.org. We collect public facts about 30 banks, transform them into 1,067 RDF triples with stable URIs, and interlink 24 of the 30 banks and all three locations with Wikidata and DBpedia through 38 owl:sameAs links, each annotated with its verification method and confidence. The dataset is described with VoID and released under CC BY-SA 4.0. A Python notebook serves as a SPARQL terminal: it materialises 3,503 additional triples with OWL RL reasoning, runs automated data quality checks, and answers 16 queries that cover the main SPARQL 1.1 query forms, including a federated query to Wikidata. We also assess the dataset against the nine Linked Data best practices.

Keywords: Linked Open Data, ontology engineering, OWL 2, SPARQL, Wikidata, DBpedia, banking.

1. Introduction

Vietnam has several dozen licensed commercial banks, and basic facts about them (legal name, ownership type, headquarters, founding year, official website) are scattered across the website of each bank, the State Bank of Vietnam, stock exchanges and encyclopedias. These facts are published for human readers. A question as simple as “which state-controlled banks founded before 1991 are headquartered in Hanoi” requires a person to open many pages and reconcile names manually, because the sources share neither identifiers nor schemas.

Linked Data addresses this problem with four principles: use URIs as names for things, use HTTP URIs so that they can be looked up, return useful information in RDF when a URI is looked up, and include links to other URIs. The five-star scheme proposed by Berners-Lee [1] grades open data from one star (available on the web under an open licence) to five stars (RDF data linked to other datasets). Large knowledge graphs of the Linked Open Data cloud [3], such as DBpedia (queryable through its public SPARQL endpoint [4]) and Wikidata, already describe the largest Vietnamese banks, but the coverage of smaller banks and of Vietnamese-language names is uneven, and no dedicated, interlinked dataset of Vietnamese banks exists.

This work builds such a dataset and an application to query it. Our contributions are:

1. an OWL 2 ontology for Vietnamese banks, bank types, locations and banking services, aligned with schema.org (Section 4);
2. a four-star RDF dataset of 30 banks with stable URIs, bilin-

- gual labels, provenance and VoID metadata (Section 5);
3. a five-star interlinking layer of 38 owl:sameAs links to Wikidata and DBpedia with a per-link evidence record (Section 6);
4. a SPARQL terminal that combines OWL reasoning, automated quality checks and 16 queries, including a federated query (Section 7), and an evaluation against Linked Data best practices (Section 8).

2. Problem Statement and Scope

2.1. Problem statement

The goal is to build a Linked Open Data application for the Vietnamese banking domain in five steps: (1) define an ontology for the domain, (2) collect relevant data, (3) transform the data to the four-star standard, (4) find and establish links to other datasets to reach five stars, and (5) provide an interface through a SPARQL endpoint or terminal to query the data.

2.2. Data policy

Only public information is used. Internal or customer data of any bank is excluded, because the first star of the scheme requires the data to be published under an open licence.

2.3. Scope options and decision

Three scopes were considered: (a) about 30 banks with basic facts and links to Wikidata and DBpedia, (b) scope (a) plus deposit interest rates by term for 10 to 15 banks, and (c) scope (b) plus branches and multi-year financial indicators. Option (c) would require crawling many differently formatted bank websites and financial statements. Given the time available,

Table 1: Minimum targets set by the group and final result.

Target	Planned	Achieved
Banks	30	30
Bank types	3 to 4	5
Locations	4 to 6	3 (see note)
Banks linked to Wikidata/DBpedia	at least 10	24
Locations linked	all	3 of 3
Source and retrieval date	every record	yes
Stable internal URIs	yes	yes

Note: the dataset has three locations because all 30 banks are headquartered in Hanoi or Ho Chi Minh City; locations with no bank are not included.

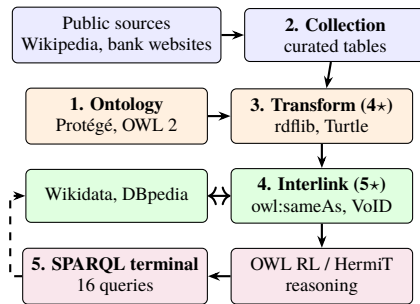


Figure 1: VNBank-LOD pipeline. The dashed arrow is the federated SERVICE query that reads Wikidata at query time.

option (a) was selected and fixed the minimum targets listed in Table 1.

3. Methodology

Figure 1 shows the pipeline. It follows the five steps of the problem: define an ontology, collect data, transform the data to four stars, link it to other datasets to reach five stars, and provide a SPARQL interface.

Tools. The ontology was edited and checked in Protégé with the HermiT reasoner. Transformation, validation and querying use Python with rdflib and the owlrl package, which implements the OWL 2 RL rule set. The query notebook runs both on Google Colab and on a local Jupyter installation.

Competency questions. We scoped the ontology with six competency questions (CQ) that the final system must answer:

- CQ1 Which banks exist and what is the type of each bank?
- CQ2 Which banks are state-owned or state-controlled?
- CQ3 Where is each bank headquartered?
- CQ4 Which banks provide a given banking service?
- CQ5 Which banks were established before 1991?
- CQ6 Which banks are linked to external knowledge graphs, and what can be retrieved through these links?

4. Ontology Design

4.1. Classes and properties

The ontology has four core classes: `:Bank`, `:BankType`, `:Location` and `:Service`. Bank types, locations and services are modelled as individuals rather than subclasses, so that new types or services can be added as data without changing the schema. Table 2 lists the properties.

Table 2: Object and datatype properties of the ontology.

Property	Kind	Domain → Range
hasBankType	object	Bank → BankType
hasHeadquarters	object	Bank → Location
isHeadquartersOf	object	inverse of hasHeadquarters
providesService	object	Bank → Service
isProvidedBy	object	inverse of providesService
operatesInCountry	object	Bank → Location
bankId	data	Bank → xsd:string
nameVi, nameEn	data	Bank → rdf:langString
abbreviation	data	Bank → xsd:string
foundedYear	data	Bank → xsd:integer
officialWebsite	data	Bank → xsd:anyURI
retrievedDate	data	Bank → xsd:date

4.2. Axioms

The class `:Bank` carries the following restrictions (Manchester syntax):

```

Bank SubClassOf
  hasBankType exactly 1 BankType,
  hasHeadquarters exactly 1 Location,
  bankId exactly 1 xsd:string,
  providesService some Service,
  operatesInCountry some Location,
  schema:BankOrCreditUnion
Bank DisjointWith BankType, Location, Service

```

Five defined classes turn competency questions into classes whose members are computed by a reasoner instead of being asserted:

```

StateOwnedBank EquivalentTo Bank and
  (hasBankType value bt:state-commercial)
JointStockBank EquivalentTo Bank and
  (hasBankType value bt:joint-stock-commercial)
ForeignBank EquivalentTo Bank and
  (hasBankType value bt:foreign-owned)
DigitalBank EquivalentTo Bank and
  (providesService value sv:digital-banking)
EstablishedBefore1991Bank EquivalentTo Bank and
  (foundedYear some xsd:integer[<= 1990])

```

The first four use `owl:hasValue` and fall inside OWL 2 RL. The last one uses a datatype restriction, which OWL 2 RL does not support; it needs a full OWL 2 DL reasoner such as HermiT (Section 7). Figure 4 shows the resulting class hierarchy in Protégé, where the reasoner also infers the inverse assertions `isProvidedBy` for each service. Figure 2 draws the same schema as a graph. It shows every class, subclass axiom, property with its domain and range, cardinality restriction and defined class of the ontology.

4.3. URI design and vocabulary reuse

All resources live under the namespace `https://data.example.org/vietnam-banking/` with one path segment per resource type: `bank/{id}`, `bank-type/{id}`, `location/{id}` and `service/{id}`. Identifiers are short lowercase slugs that do not change when a bank is renamed; for example `bank/lpbank` remains stable although the bank was renamed from `LienVietPostBank` to `LPBank` in 2024. Every bank also carries `rdfs:label` and `schema:name` in Vietnamese and English and `schema:url`, so that generic Linked Data browsers can display it. Dataset metadata uses Dublin Core terms and `Void`.

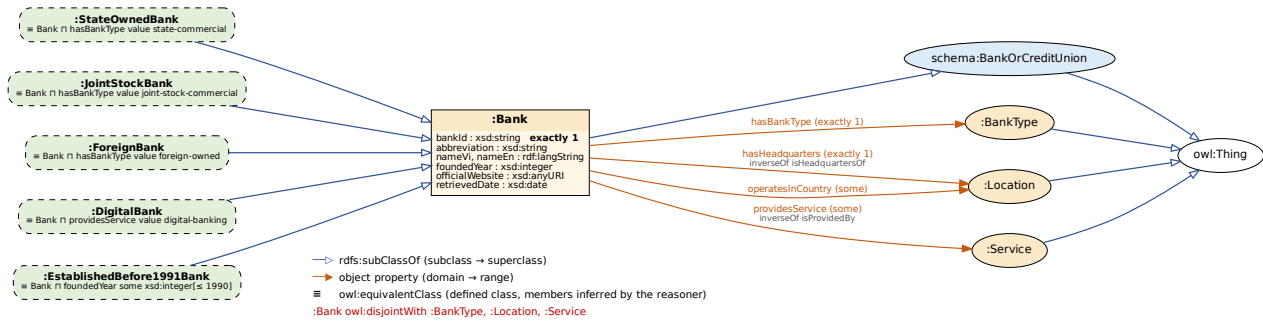


Figure 2: Ontology schema. Blue hollow arrows are `rdfs:subClassOf` (subclass to superclass), orange arrows are object properties from domain to range with their cardinality and inverse, and green dashed boxes are the five defined classes with their `owl:equivalentClass` definitions. `:BankType`, `:Location` and `:Service` sit directly under `owl:Thing` and are disjoint with `:Bank`.

5. Data Collection and Transformation

5.1. Sources and scope

The dataset covers 30 banks: two policy banks, five state-owned or state-controlled commercial banks, twenty joint-stock commercial banks, two domestic single-member banks owned by a joint-stock bank after compulsory transfer (MBV and GPBank), and one wholly foreign-owned bank. Names, headquarters and websites come from the Wikipedia list of banks in Vietnam (CC BY-SA 4.0) and were checked against the official website of each bank. Each bank is annotated with `dcterms:source` and `retrievedDate`.

Two domain issues required manual curation. First, the administrative reform of 1 July 2025 removed the district level in Vietnam, so a bank cannot be placed in a former district-level city; all headquarters are mapped to provincial-level units. Second, several banks changed legal form or name in 2024–2025 after compulsory transfer, so we record the current legal name and keep the bank identifier stable.

5.2. Transformation to four stars

The curated tables were transformed with `rdflib`. Each row becomes one resource; each column becomes a typed literal or a link to a type, location or service individual. Listing 1 shows the resulting description of one bank.

```
bank:acb a :Bank ;
  rdfs:label "Ngân hàng TMCP Á Châu"@vi,
    "Asia Commercial Joint Stock Bank"@en ;
  :bankId "ACB" ; :foundedYear 1993 ;
  :hasBankType bt:joint-stock-commercial ;
  :hasHeadquarters loc:ho-chi-minh-city ;
  :operatesInCountry loc:vietnam ;
  :providesService sv:card, sv:digital-banking ;
  :officialWebsite "https://www.acb.com.vn/"^^xsd:anyURI ;
  schema:url <https://www.acb.com.vn/> ;
  dcterms:source <https://en.wikipedia.org/wiki/List_of_banks_in_Vietnam> ;
  owl:sameAs <http://www.wikidata.org/entity/Q727019>,
    <http://dbpedia.org/resource/Asia_Commercial_Bank> .
```

Listing 1: Turtle description of ACB (abridged).

Table 3 summarises the dataset. All tables are also exported from the same Turtle file to CSV and Excel, so the tabular and RDF views cannot diverge. Figure 3 shows the resulting graph around ACB and Vietcombank, and Figure 5

Table 3: Dataset statistics.

Item	Count
Asserted triples	1,067
Triples after OWL RL reasoning	4,570
Banks	30
Bank types	5
Locations (all used and linked)	3
Service categories	9
Bank to service assertions	202
owl:sameAs links (banks / locations)	32 / 6
Banks with at least one external link	24 / 30

Table 4: Banks per type (query Q9).

Bank type	Banks
Joint-stock commercial bank	20
State-owned or state-controlled commercial bank	5
Privately owned domestic commercial bank	2
Policy bank	2
Wholly foreign-owned bank	1
Total	30

shows how all 30 banks connect to the five bank types and the two headquarters cities.

5.3. Bank types

Table 4 gives the distribution of banks by type, as returned by an aggregate SPARQL query (Q9).

6. Interlinking to Five Stars

6.1. Linking strategy

We link internal resources to Wikidata and DBpedia with `owl:sameAs`. Three verification methods were used, and each link is stored with its method, an evidence URL and a confidence level in `external_links.csv`:

- DBpedia page check* (high confidence): the DBpedia page was opened and its description confirmed to be the Vietnamese bank; the Wikidata IRI was then read from the `owl:sameAs` statements of that page.
- Wikidata search* (medium or low confidence): the bank name was searched on `wikidata.org` and the item whose label matched the bank was selected.

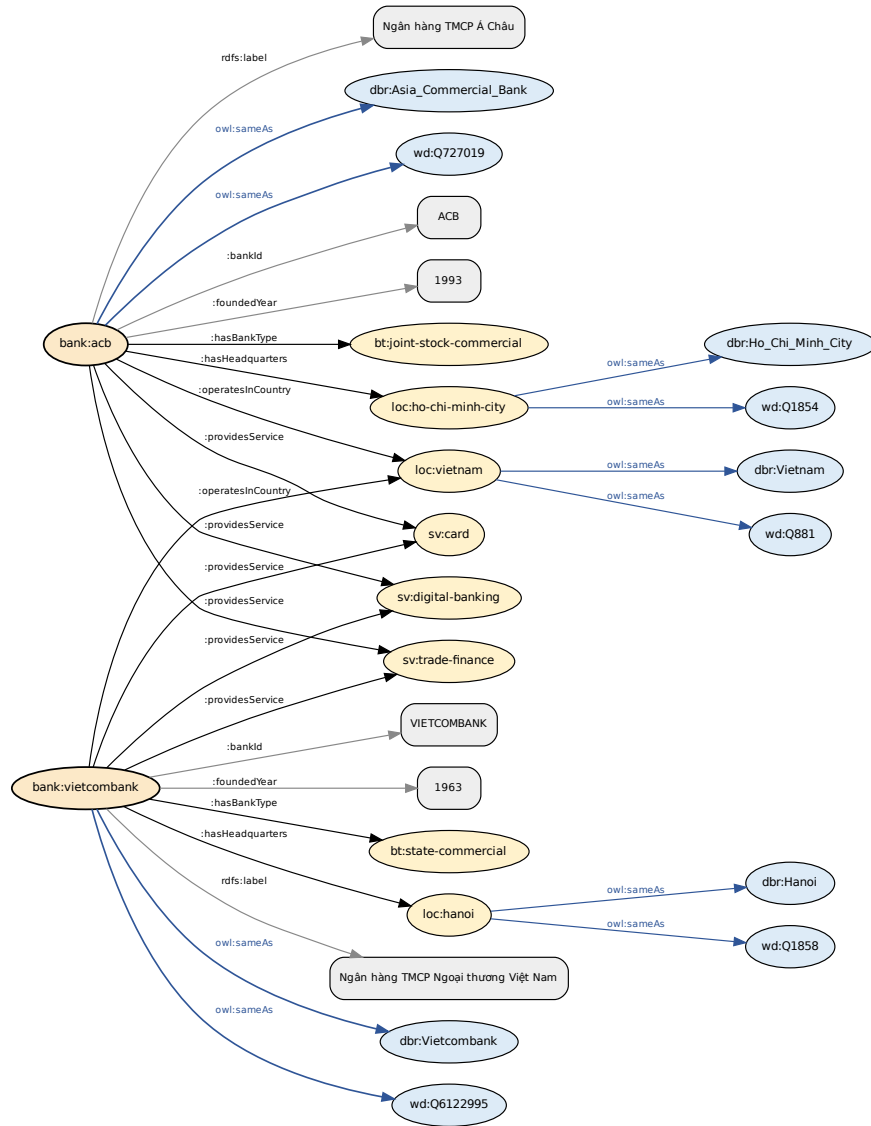


Figure 3: Instance subgraph of ACB and Vietcombank, generated from the same file: each bank with its type, headquarters, three of its services, two literals and its owl:sameAs links (blue) to Wikidata and DBpedia.

Table 5: External links by target, method and confidence.

Subject	Method	Conf.	Links
Bank	Wikidata search	medium	18
Bank	Wikidata search	low	1
Bank	DBpedia page check	high	6
Bank	Wikidata IRI from DBpedia	high	5
Bank	Wikipedia title	medium	2
Location	Wikidata search	high	3
Location	DBpedia page / title	high	3
Total			38

- *Wikipedia title* (medium confidence): the DBpedia IRI was derived from an existing English Wikipedia article title. Table 5 summarises the result. The links are declared as two void:Linkset resources, one per target dataset, whose void:triples values match the actual number of links.

6.2. Effect of owl:sameAs on queries

owl:sameAs states that two IRIs denote the same individual, so a reasoner copies every property in both directions. After OWL RL reasoning, an internal bank therefore has three equivalent IRIs, and a naive query that returns the headquarters IRI lists each bank three times. Our queries either match the internal IRI directly or filter on the internal namespace. The same effect explains why the Protégé screenshots in Figure 6 list Wikidata and DBpedia IRIs next to internal banks.

7. SPARQL Terminal and Reasoning

7.1. Notebook architecture

The notebook loads the Turtle file into two graphs: `g_raw` with the asserted triples, and `g` with the OWL RL closure. Statistics, link listings, CONSTRUCT and DESCRIBE run on `g_raw`; queries over defined classes and inverse properties run on `g`. A function `sparql()` accepts any query and returns

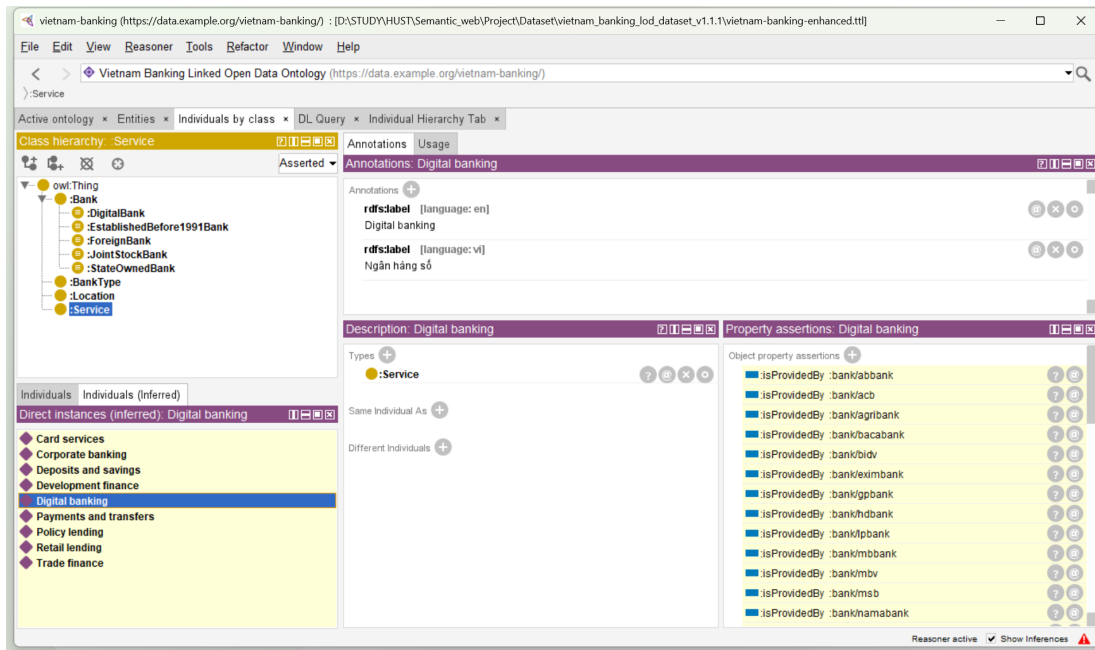


Figure 4: Inferred class hierarchy in Protégé. The five defined classes are classified under `:Bank`; the right panel shows `isProvidedBy` assertions inferred from `owl:inverseOf`.

Table 6: Queries of the SPARQL terminal. Q15 depends on the live Wikidata endpoint; its result size is not fixed.

ID	Question	SPARQL features	CQ	Result
Q1	All banks with names and type	SELECT, OPTIONAL, property path, ORDER BY	CQ1	30 rows
Q2	State-owned or state-controlled banks	inferred class <code>StateOwnedBank</code>	CQ2	5 rows
Q3	Banks headquartered in Hanoi	match on internal IRI	CQ3	20 rows
Q4	Banks providing digital banking	inferred class <code>DigitalBank</code>	CQ4	28 rows
Q5	Banks founded in or before 1990	numeric FILTER	CQ5	6 rows
Q6	External links of each bank	BIND, IF	CQ6	32 rows
Q7	Service to bank (inverse)	inferred <code>isProvidedBy</code>	CQ4	28 rows
Q8	Headquarters to bank (inverse)	inferred <code>isHeadquartersOf</code>	CQ3	20 rows
Q9	Number of banks per type	COUNT, GROUP BY	CQ1	5 rows
Q10	Does the dataset have five-star links?	ASK	CQ6	true
Q11	Bank and service graph in schema.org terms	CONSTRUCT	CQ4	280 triples
Q12	Banks without external links	FILTER NOT EXISTS	CQ6	6 rows
Q13	State-owned or foreign banks	UNION, BIND	CQ2	6 rows
Q14	Full description of ACB	DESCRIBE	CQ1	27 triples
Q15	Wikidata label and inception of linked banks	SERVICE (federated), VALUES	CQ6	live
Q16	Banks per headquarters city with DBpedia link	free query in terminal, GROUP BY	CQ3	2 rows

a table, so the notebook serves as a SPARQL terminal. Every result is exported to CSV or Turtle.

7.2. Reasoning results

OWL RL reasoning adds 3,503 triples to the 1,067 asserted ones and produces no member of `owl:Nothing`, so the data is consistent with the ontology. Among internal resources, the reasoner classifies 5 banks as `StateOwnedBank`, 20 as `JointStockBank`, 1 as `ForeignBank` and 28 as `DigitalBank`, and it derives 202 `isProvidedBy` and 30 `isHeadquartersOf` assertions. `EstablishedBefore1991Bank` receives no member under OWL RL because of its datatype restriction. Hermit in Protégé evaluates the same class expression and returns the six expected banks (Figure 6c), which matches the FILTER-based query Q5.

7.3. Data quality checks

Before querying, the notebook runs SPARQL checks derived from the ontology axioms (Table 7). They detect violations that an open-world reasoner does not report, such as a bank with no type: OWL treats a missing value as unknown, not as an error. The final dataset passes all checks.

Table 7: Automated data quality checks.

Check	Violations
Bank without bank type	0
Bank without headquarters	0
Bank with more than one type	0
Bank with more than one headquarters	0
Bank without <code>rdfs:label</code>	0
Location not used by any bank	0
Members of <code>owl:Nothing</code> after reasoning	0

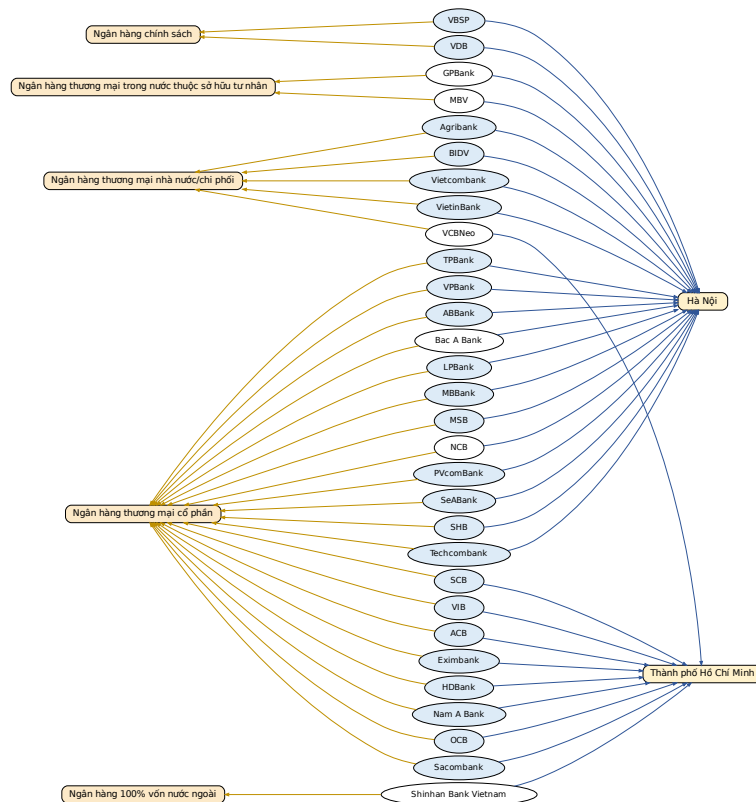


Figure 5: Overview of the 30 banks: each bank (centre) is connected to its bank type (left) and its headquarters (right). Shaded banks have at least one owl:sameAs link to Wikidata or DBpedia; white banks have none. Generated from the Turtle file by make_kg.py.

7.4. Queries

Table 6 lists the 16 queries, the SPARQL features they exercise, the competency question they answer and their result size.

Figure 7 shows four cells of the executed notebook, including the quality checks and the aggregate query Q9. Listing 2 shows Q3. Matching the internal IRI loc:hanoi rather than a label avoids the duplication caused by owl:sameAs; the inverse query Q8 returns the same 20 banks, which cross-checks the inverse property axiom.

```
SELECT ?id ?nameVi WHERE {
  ?bank :hasHeadquarters loc:hanoi ;
        :bankId ?id ; :nameVi ?nameVi .
  FILTER(STRSTARTS(STR(?bank),
    "https://data.example.org/vietnam-banking/bank/"))
} ORDER BY ?nameVi
```

Listing 2: Q3: banks headquartered in Hanoi.

Listing 3 shows the federated query. It sends the Wikidata IRIs of the linked banks to the Wikidata endpoint and returns, for each bank, the English label and the inception date (property P571) next to the founding year stored locally. This query is the operational proof of the fifth star: the links are used at query time to retrieve data that the local dataset does not contain.

```
SELECT ?id ?foundedYear ?wdLabel ?wdInception WHERE {
  ?bank :bankId ?id ; :foundedYear ?foundedYear ;
        owl:sameAs ?wd .
  SERVICE <https://query.wikidata.org/sparql> {
    VALUES ?wd { wd:Q727019 wd:Q6122995 ... }
    ?wd rdfs:label ?wdLabel .
    FILTER(LANG(?wdLabel) = "en")
    OPTIONAL { ?wd wdt:P571 ?wdInception }
  }
}
```

Listing 3: Q15: federated query to Wikidata (abridged).

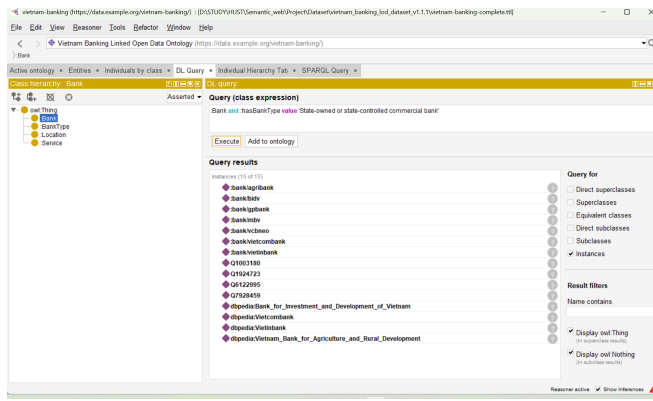
8. Evaluation

8.1. Five-star scheme

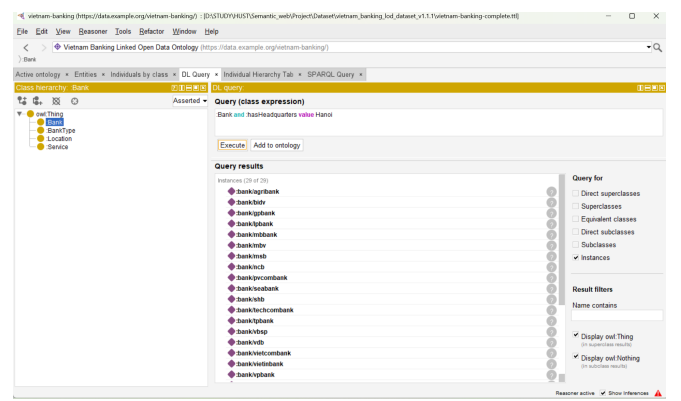
The dataset is available on the web under an open licence (CC BY-SA 4.0, one star), as structured data (two stars) in a non-proprietary format (Turtle and CSV, three stars), using W3C standards with a URI for every resource (RDF, RDFS, OWL, SPARQL, four stars), and linked to Wikidata and DBpedia (five stars).

8.2. Linked Data best practices

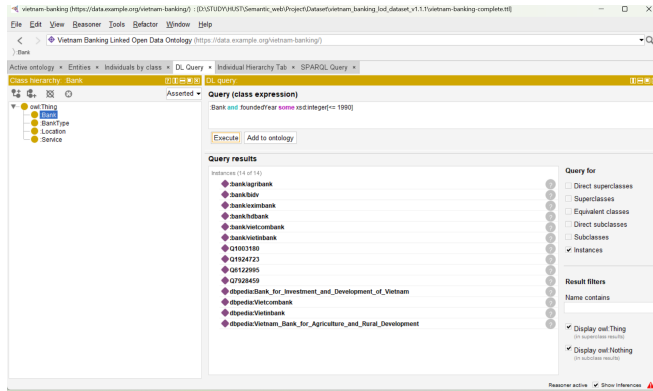
Table 8 assesses the dataset against the nine best practices of Heath and Bizer. Six are fully met, two are partly met, and one is not met.



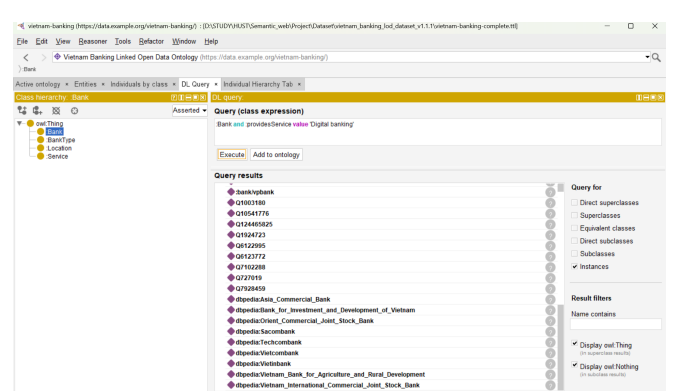
(a) Bank and hasBankType value state-commercial



(b) Bank and hasHeadquarters value Hanoi



(c) Bank and foundedYear some xsd:integer[<= 1990]



(d) Bank and providesService value Digital banking

Figure 6: DL queries answered by HermiT in Protégé. Result lists include Wikidata and DBpedia IRIs because they are owl:sameAs-equivalent to internal banks. The screenshots predate the reclassification of MBV and GPBank; on the final dataset, query (a) returns five internal banks.

Table 8: Compliance with Linked Data best practices.

Best practice	Status
1. Provide dereferenceable URIs	partly (placeholder domain)
2. Set RDF links to other sources	met (38 links)
3. Use widely deployed vocabularies	met (schema.org, DC, VoID)
4. Make own terms dereferenceable	not met
5. Map own terms to other vocabularies	partly (Bank $\sqsubseteq$ schema)
6. Provide provenance metadata	met (source, link evidence)
7. Provide licensing metadata	met (dcterms:license)
8. Provide dataset-level metadata	met (VoID)
9. Refer to additional access methods	met (data dump, notebook)

8.3. Limitations

First, bank to service assertions are assigned by bank category rather than checked bank by bank, so the 28 commercial banks share the same seven services and Q4 does not discriminate between them. Second, 19 of the 32 bank links were selected by label matching on Wikidata and carry medium or low confidence. Third, the namespace data.example.org is a placeholder, so URIs cannot yet be dereferenced, and no public SPARQL endpoint is deployed. Fourth, the dataset holds about ten attributes per bank and only three locations, because all 30 banks are headquartered in Hanoi or Ho Chi Minh City. Finally, six banks have no external link (Q12),

mostly banks renamed recently whose Wikidata items we could not confirm.

9. Conclusion and Future Work

VNBANK-LOD shows that a small team can move public information about Vietnamese banks from one or two stars to five stars with standard Semantic Web tools. The OWL 2 ontology answers the six competency questions through a combination of asserted data, OWL RL and HermiT reasoning, and the SPARQL terminal demonstrates every query form of SPARQL 1.1, including federation with Wikidata.

Future work includes publishing the dataset under a persistent domain with content negotiation and an Apache Jena Fuseki endpoint, enriching each bank with SWIFT codes, stock tickers, former names and charter capital, checking services against bank websites, aligning the ontology with FIBO, and replacing manual matching with link discovery frameworks such as Silk [2].

Project Repository

The whole project is publicly available at:

<https://github.com/Huongdinh91/vietnam-banking-lod>

The repository holds the Turtle file vietnam-banking.ttl, the CSV and Excel exports, the executed notebook, all query results, the data build script, the script that draws the knowledge graph figures, a README and a change log.

References

- [1] M. Hausenblas and J. G. Kim. *5-star Open Data*. <https://5stardata.info/en/>
- [2] *Silk: The Linked Data Integration Framework*. <http://silkframework.org/>
- [3] J. P. McCrae. *The Linked Open Data Cloud*. Insight Centre for Data Analytics. <https://lod-cloud.net/>
- [4] DBpedia Association. *SPARQL over Online Databases*. <https://www.dbpedia.org/resources/sparql/>

In [5]:

```
# Kiểm tra ràng buộc của ontology trên dữ liệu gốc
checks = {
    "Ngân hàng": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank a :Bank . FILTER(STRSTARTS(STR(?bank), "https://data.
example.org/vietnam-banking/bank/")) }''',
    "Thiếu loại ngân hàng": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank a :Bank . FILTER(STRSTARTS(STR(?bank), "ht
tps://data.example.org/vietnam-banking/bank/")) FILTER NOT EXISTS { ?bank :hasBankType ?t } }''',
    "Thiếu trụ sở": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank a :Bank . FILTER(STRSTARTS(STR(?bank), "https://da
ta.example.org/vietnam-banking/bank/")) FILTER NOT EXISTS { ?bank :hasHeadquarters ?h } }''',
    "Nhiều hơn 1 loại": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank :hasBankType ?a, ?b . FILTER(?a != ?b) }''',
    "Nhiều hơn 1 trụ sở": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank :hasHeadquarters ?a, ?b . FILTER(?a != ?b)
}''',
    "Thiếu nhãn rdfs:label": '''SELECT (COUNT(DISTINCT ?bank) AS ?n) WHERE { ?bank a :Bank . FILTER(STRSTARTS(STR(?bank), "h
ttps://data.example.org/vietnam-banking/bank/")) FILTER NOT EXISTS { ?bank rdfs:label ?l } }''',
    "Địa điểm không được dùng": '''SELECT (COUNT(DISTINCT ?l) AS ?n) WHERE { ?l a :Location . FILTER NOT EXISTS { ?x :hasHea
dquarters|:operatesInCountry ?l } }''',
}
quality = pd.DataFrame([(k, int(sparql(q, g_raw)[0]) for k, q in checks.items()), columns=["Kiểm tra", "Số lượng"])
display(quality)
```

	Kiểm tra	Số lượng
0	Ngân hàng	30
1	Thiếu loại ngân hàng	0
2	Thiếu trụ sở	0
3	Nhiều hơn 1 loại	0
4	Nhiều hơn 1 trụ sở	0
5	Thiếu nhãn rdfs:label	0
6	Địa điểm không được dùng	0

(a) Data quality checks

In [8]:

```
df3 = run_query(
    "TRUY VẤN 3: NGÂN HÀNG CÓ TRỤ SỞ TẠI HÀ NỘI",
    """
    SELECT ?id ?nameVi
    WHERE {
        ?bank :hasHeadquarters loc:hanoi ;
        :bankId ?id ;
        :nameVi ?nameVi .
        FILTER(STRSTARTS(STR(?bank), "https://data.example.org/vietnam-banking/bank/"))
    }
    ORDER BY ?nameVi
    """,
    "query_03_hanoi_headquarters.csv",
    graph=g,
)
```

TRUY VẤN 3: NGÂN HÀNG CÓ TRỤ SỞ TẠI HÀ NỘI
Số kết quả: 20

	id	nameVi
0	VBSP	Ngân hàng Chính sách Xã hội Việt Nam
1	AGRIBANK	Ngân hàng Nông nghiệp và Phát triển Nông thôn Việt Nam
2	VDB	Ngân hàng Phát triển Việt Nam
3	ABBBANK	Ngân hàng TMCP An Bình
4	BACABANK	Ngân hàng TMCP Bắc Á
5	VIETINBANK	Ngân hàng TMCP Công thương Việt Nam
6	MSB	Ngân hàng TMCP Hàng Hải Việt Nam
7	TECHCOMBANK	Ngân hàng TMCP Kỹ thương Việt Nam

(b) Q3, first rows of the 20 results

Figure 7: Screenshots of cells from the executed notebook Group_24_Semantic_Web_project_final.ipynb.

In [14]:

```
df9 = run_query(
    "TRUY VẤN 9: SỐ NGÂN HÀNG THEO LOẠI",
    """
    SELECT ?bankType (COUNT(DISTINCT ?bank) AS ?numberOfBanks)
    WHERE {
        ?bank a :Bank ;
              :hasBankType/rdfs:label ?bankType .
        FILTER(LANG(?bankType) = "vi")
        FILTER(STRSTARTS(STR(?bank), "https://data.example.org/vietnam-banking/bank/"))
    }
    GROUP BY ?bankType
    ORDER BY DESC(?numberOfBanks)
    """,
    "query_09_count_banks_by_type.csv",
    graph=g_raw,
)
```

TRUY VẤN 9: SỐ NGÂN HÀNG THEO LOẠI
Số kết quả: 5

	bankType	numberOfBanks
0	Ngân hàng thương mại cổ phần	20
1	Ngân hàng thương mại nhà nước/chi phối	5
2	Ngân hàng thương mại trong nước thuộc sở hữu tư nhân	2
3	Ngân hàng chính sách	2
4	Ngân hàng 100% vốn nước ngoài	1

(c) Q9, aggregation by bank type

In [21]:

```
wikidata_iris = sorted([
    str(o) for s, o in g_raw.subject_objects(OWL.sameAs)
    if str(s).startswith("https://data.example.org/vietnam-banking/bank/")
    and str(o).startswith("http://www.wikidata.org/entity/")
])
values = " ".join(f"<{iri}>" for iri in wikidata_iris)
print("Số IRI Wikidata gửi đi:", len(wikidata_iris))

query_15 = f"""
SELECT ?id ?nameVi ?foundedYear ?wikidata ?wikidataLabel ?wikidataInception
WHERE {{
    ?bank a :Bank ;
          :bankId ?id ;
          :nameVi ?nameVi ;
          :foundedYear ?foundedYear ;
          owl:sameAs ?wikidata .
    FILTER(STRSTARTS(STR(?bank), "https://data.example.org/vietnam-banking/bank/"))
    SERVICE <https://query.wikidata.org/sparql> {{
        VALUES ?wikidata {{ {values} }}
        ?wikidata rdfs:label ?wikidataLabel .
        FILTER(LANG(?wikidataLabel) = "en")
        OPTIONAL {{ ?wikidata wdt:P571 ?wikidataInception }}
    }}
}}
ORDER BY ?nameVi
"""

try:
    df15 = run_query("TRUY VẤN 15: BỔ SUNG DỮ LIỆU TỪ WIKIDATA", query_15,
                     "query_15_wikidata_service.csv", graph=g_raw)
except Exception as e:
    df15 = None
    print("Không truy cập được Wikidata từ môi trường hiện tại:", type(e).__name__)
    print("Hãy chạy lại ô này khi có Internet (ví dụ trên Google Colab).")
```

(d) Q15, federated query to Wikidata (code)

Figure 7: Screenshots of cells from the executed notebook Group_24_Semantic_Web_project_final.ipynb. (continued)